

24.2 PET Scanning

Question Paper

Course	CIEA Level Physics
Section	24. Medical Physics
Topic	24.2 PET Scanning
Difficulty	Hard

Time allowed: 60

Score: /50

Percentage: /100

Question 1a

Positron Emission Tomography (PET) and ultrasound scans are both used in medical diagnosis.

Compare the effectiveness of these scans in terms of

(i)
the images and the information that can be obtained,

[6]

(ii)
patient safety.

[4]

[10 marks]

Question 1b

Table 1.1 gives the properties of two radionuclides.

Table 1.1

	Rubidium 82	Iodine 124
half-life	75 sec	4.18 days
fraction of β^+ emitted	0.95	0.22
radiation emitted	β^+, γ	β^+, γ

By considering the information in Table 1.1, suggest which of these nuclides is more suitable for use as a tracer in a PET scan.

[4 marks]

Question 2a

Fluorodeoxyglucose (FDG) is a radiopharmaceutical used for PET scans. It contains radioactive fluorine-18 (F-18), which decays by positron emission with a half-life of 110 minutes.

In preparation for a scan scheduled for 12:00 p.m., a radiologist prepares a sample of F-18 FDG at 5 a.m. earlier in the day. For this scan, the radiologist determines the optimum activity of fluorine-18 would be 10 mCi when the scan starts at 12:00 p.m.

Determine the activity, in GBq, of the dose of fluorine-18 that the radiologist should prepare at 5 a.m.

$$1 \text{ mCi} = 3.7 \times 10^7 \text{ Bq}$$

[3 marks]

Question 2b

The patient is scheduled to come in an hour earlier, at 11:00 a.m. to receive their injection of F-18 FDG.

At this facility, the maximum activity limit of F-18 FDG that can be injected into patients is set at 14 mCi.

The radiologist forgets to factor this into their calculations when preparing the dose of F-18 FDG and discovers that it has an activity of 15.3 mCi at 11:00 a.m. when the patient arrives.

Determine the earliest time, to the nearest minute, at which the sample of F-18 FDG may be injected into the patient.

[2 marks]

Question 2c

About 9.9% of the mass of F-18 FDG is fluorine-18. The molar mass of fluorine-18 is $0.018 \text{ kg mol}^{-1}$.

(i)

Calculate the initial mass, in kg, of F-18 FDG given to the patient at the time of injection.

[3]

(ii)

Determine the fraction of F-18 FDG remaining in the patient's body 6 hours after the initial injection.

[2]

(iii)

Suggest why the actual fraction would likely be lower than the value you calculated in (c)(i).

[1]

[6 marks]

Question 2d

Both computed tomography (CT) scans and positron emission tomography (PET) scans can produce detailed pictures of the organs and tissues inside the body. Both techniques share many similarities and differences.

In recent years, it has become much more common for doctors to use a PET-CT scan which is a machine able to perform both scans simultaneously. This way, doctors can combine the images from both scans to create an enhanced picture of the tissue or organ being studied.

(i)

Describe the major differences between a CT scan and a PET scan.

[4]

(ii)

Discuss the benefits and risks of using a PET-CT scanner instead of using either scan separately in medical diagnosis.

[2]

[6 marks]

Question 3a

Table 1.1 shows some information about two radionuclides commonly used in PET scanning.

Table 1.1

	Fluorine 18	Oxygen 15
half-life	109.8 min	122 s
maximum positron energy / MeV	0.63	1.73
% of decays that occur by β^+	97%	100%

Under certain conditions, a gamma photon may be converted into an electron and a positron.

Deduce, with suitable calculations, if the decay of one or both radionuclides in Table 1.1 can result in both the annihilation and the creation of electron-positron pairs.

[4 marks]

Question 3b

In practice, it is observed that the majority of gamma photons emitted have an energy of 511 keV and travel at exactly 180° to each other, regardless of the initial energy of the positron.

When fluorine-18 is used as a tracer, a very small proportion of the gamma photons have energies greater than 511 keV and do not travel at exactly 180° to each other.

When oxygen-15 is used as a tracer, a slightly larger proportion of the gamma photons have energies greater than 511 keV and do not travel at exactly 180° to each other.

(i)

Explain these observations.

[6]

(ii)

Suggest a potential issue in the detection of the gamma rays if the majority of photons possessed energies greater than 511 keV.

[2]

[8 marks]**Question 3c**

A tracer contains radioactive fluorine-18 and has an initial activity of 450 MBq. A typical PET scan using fluorine-18 takes 30 minutes.

Determine the number of photons produced during a PET scan using fluorine-18. Assume that all photons produced in annihilation events are detected by the detector.

[4 marks]

Question 3d

A typical PET scan using oxygen-15 takes 6 minutes.

Determine the initial activity of oxygen-15 required to produce the same number of photons as the scan using fluorine-18.

[3 marks]



Model Answers: Hard

1a

(a)

(i) Comparison of the images and the information obtained from PET and ultrasound:

From a PET scan:

Any **three** from:

- Able to detect real-time metabolic changes / cellular activity; [1 mark]
- Tissue which requires more energy takes up / absorbs more of the tracer e.g. cancerous tissue; [1 mark]
- Able to detect activity / abnormalities in brain / heart / organ function; [1 mark]
- Can provide quick and detailed information regarding cancer malignancy and tumour spreading; [1 mark]

From an ultrasound scan:

Any **three** from:

- Able to image the size, position and real-time movement of a foetus / soft tissues / organs; [1 mark]
- Cannot penetrate bone, hence cannot image broken bones / examine brain tissue; [1 mark]
- Cannot pass through air spaces / lungs; [1 mark]
- Cannot distinguish between tissues with very different (acoustic) impedances e.g. benign and malignant solid masses; [1 mark]

(ii) Comparison of patient safety in PET and ultrasound scans:

Safety in PET scanning:

Any **two** from:

- Non-invasive procedure; [1 mark]
- (However) requires the injection of a (weakly) radioactive nuclide; [1 mark]
- The radiation exposure to patients is moderate to high; [1 mark]
- Can only be used when the benefits of scan outweigh the risks of radiation; [1 mark]

Safety in ultrasound scanning:

Any **two** from:

- Non-invasive procedure; [1 mark]
- No exposure to ionising radiation / tracer; [1 mark]
- No need for injection of contrast dye or tracer; [1 mark]
- The only scanning technique that is 100% safe for imaging a foetus; [1 mark]

[Total: 10 marks]

1b

(b) Compare the two radionuclides for use in a PET scan:

Any **three** from:

- Rb-82 emits higher proportion of β^+ which means it will cause more annihilation events (compared to I-124); [1 mark]
- I-124 emits higher proportion of gamma (by decay) which could interfere with the detection of gamma photons from annihilation events; [1 mark]
- Half-life of 4.18 days means I-124 will remain radioactive in the body for longer hence more likely to damage cells / increase radiation dose for patient (and others they are in contact with); [1 mark]
- Half-life of 75 sec means Rb-82 may decay too quickly for certain types of diagnosis to be undertaken; [1 mark]

Write a sensible conclusion based on the above points:

- Rb-82 is the better isotope because it has a short half-life, so it produces a high activity for a short period but does not remain in the body for long; [1 mark]
- OR**
- I-124 may be the better isotope depending on the organ / tissue being studied / better if a longer scan is required / computer software should be able to identify

if gamma is from decay or annihilation event; [1 mark]

[Total: 4 marks]

Remember that gamma rays are not only just produced through annihilation, but also from radioactive substances decaying by emitting gamma rays. This means that when 0.22 of the I-124 decays with beta plus, 0.78 decays by gamma radiation. This is a lot more than the Rb-82 which only has 0.5 of its decay by gamma.

2a

(a) Determine the activity of the dose of fluorine-18 that the radiologist should prepare at 5 a.m.:

List the known quantities:

- Half-life, $t_{1/2} = 110$ min
- Time, $t = 7$ hours $= 7 \times 60 = 420$ min
- Optimum activity, $A = 10$ mCi
- $1 \text{ mCi} = 3.7 \times 10^7 \text{ Bq}$

Determine the initial activity of the dose of F-18:

- Exponential decay: $A = A_0 e^{-\lambda t} \Rightarrow A_0 = \frac{A}{e^{-\lambda t}}$

- Decay constant: $\lambda = \frac{\ln 2}{t_{1/2}}$

- $A_0 = \frac{A}{e^{-\frac{t \ln 2}{t_{1/2}}}} = \frac{10}{e^{-\frac{420 \times \ln 2}{110}}} \text{ [1 mark]}$

- $A_0 = 141.05 \text{ mCi [1 mark]}$

Convert the initial activity to GBq:

- $A_0 = 141 \times (3.7 \times 10^7) = 5.219 \times 10^9 \text{ Bq}$

- $A_0 = 5.2 \text{ GBq [1 mark]}$

[Total: 3 marks]

The Curie (Ci) was the original unit used to express the decay rate (activity) of a sample of radioactive material. Since 1 Ci is equal to 37 GBq (giga-becquerels), it is a more convenient unit to use, particularly in medical physics where decay rates of common radionuclides are of the order MBq and GBq

2b

(b) Determine the time at which the sample of F-18 FDG may be injected into the patient:

List the known quantities:

- Initial activity (at 11 a.m.), $A_0 = 15.3 \text{ mCi}$
- Maximum activity, $A = 14 \text{ mCi}$
- Decay constant, $\lambda = 6.3 \times 10^{-3} \text{ min}^{-1}$

Determine the time taken to reach the required activity level:

- Exponential decay: $A = A_0 e^{-\lambda t} \Rightarrow \frac{A}{A_0} = e^{-\lambda t}$

- $\frac{A}{A_0} = e^{-\lambda t} \Rightarrow \ln\left(\frac{A}{A_0}\right) = -\lambda t$

- $t = -\frac{1}{\lambda} \ln\left(\frac{A}{A_0}\right) \Rightarrow t = \frac{\ln\left(\frac{A_0}{A}\right)}{\lambda}$

- $t = \frac{\ln\left(\frac{15.3}{14}\right)}{6.3 \times 10^{-3}} = 14.09 \text{ min}$ [1 mark]

- Earliest time the sample can be injected: 11:15 a.m. [1 mark]

[Total: 2 marks]

Technically the earliest time would be 11 am + 14.09 minutes which would be 11:14 am and 5.4 seconds! However, since we wouldn't want a time that precise, a sensible time of 11:15 am is required for the final answer.

$$A_0 \quad \left(A_0 \right)$$

$$\bullet \quad t = -\frac{1}{\lambda} \ln \left(\frac{A}{A_0} \right) \Rightarrow t = \frac{\ln \left(\frac{A_0}{A} \right)}{\lambda}$$

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- Earliest time the sample can be injected: 11:15 a.m. [1 mark]

[Total: 2 marks]

Technically the earliest time would be 11 am + 14.09 minutes which would be 11:14 am and 5.4 seconds! However, since we wouldn't want a time that precise, a sensible time of 11:15 am is required for the final answer.

2c

(c)

List the known quantities:

- Activity of F-18 at injection, $A_0 = 14 \text{ mCi} = 14 \times (3.7 \times 10^7) \text{ Bq}$
- Decay constant, $\lambda = 6.3 \times 10^{-3} \text{ min}^{-1}$ or $1.05 \times 10^{-4} \text{ s}^{-1}$
- Molar mass of F-18, $M_r = 0.018 \text{ kg mol}^{-1}$
- Avogadro's constant, $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$

(i) Determine the initial number of F-18 nuclei in the injected sample:

$$\bullet \quad \text{Activity: } A = \lambda N \Rightarrow N_0 = \frac{A_0}{\lambda}$$

$$\bullet \quad \text{Initial number of nuclei: } N_0 = \frac{14 \times (3.7 \times 10^7)}{1.05 \times 10^{-4}}$$

$$\bullet \quad \text{Initial number of nuclei: } N_0 = 4.93 \times 10^{12} \text{ [1 mark]}$$

Use the equation relating mass, moles and molar mass:

- Moles of a substance: $n = \frac{N}{N_A}$
- Mass of a substance: $m = n \times M_r \Rightarrow m = \frac{N_0}{N_A} \times M_r$
- Mass of F-18: $m = \frac{4.93 \times 10^{12}}{6.02 \times 10^{23}} \times 0.018 = 1.47 \times 10^{-13} \text{ kg}$ [1 mark]
- Mass of F-18 FDG: $\frac{1.47 \times 10^{-13}}{0.099} = 1.5 \times 10^{-12} \text{ kg}$ [1 mark]

(ii) The fraction of F-18 FDG remaining in the patient's body after 6 hours:

- Exponential decay: $N = N_0 e^{-\lambda t} \Rightarrow \frac{N}{N_0} = e^{-\lambda t}$
- Fraction of F-18 remaining: $\frac{N}{N_0} = e^{-(6.03 \times 10^{-3}) \times 6 \times 60}$
- Fraction of F-18 remaining: $\frac{N}{N_0} = 0.11$ [1 mark]
- Fraction of F-18 FDG remaining: $\frac{m}{m_0} = 1 - 0.11 = 0.89$ [1 mark]

(iii) The actual fraction would likely be lower than 0.89 because:

- The patient would flush out / excrete the F-18 FDG through bodily functions / biological means; [1 mark]

[Total: 6 marks]

2d

(d)

(i) The major differences between a CT scan and a PET scan are:

Any **two** pairs from:

The materials used:

- In a CT scan: a contrast dye is injected into the patient and X-rays are passed through the body; [1 mark]
- In a PET scan: a radioactive tracer / positron-emitting substance is injected into the patient; [1 mark]

The methods of detecting radiation:

- In a CT scan: the patient is surrounded by (an array of) X-ray detectors which analyse the intensity of X-rays after passing through the patient; [1 mark]
- In a PET scan: the patient is surrounded by (a ring of) gamma detectors which analyse the difference in time of arrival of the two gamma photons created by the annihilation of a positron (from the tracer) and electron (from the patient); [1 mark]

The images they produce / information they provide:

- CT scans produce: a detailed three-dimensional image made up of many images / slices of a region at different angles (so it can be rotated), can be used for hard tissues (bones) and soft tissues (heart, brain etc); [1 mark]
- PET scans produce: real-time (three-dimensional) images of metabolic / cellular activity in a region of the body, particularly useful for detecting tumours and diagnosing neurological conditions; [1 mark]

The amount of radiation present in the patient after the scan:

- In a CT scan: no radiation remains in the body; [1 mark]
- In a PET scan: a small amount of radiation remains in the body for a short time; [1 mark]

(ii) The benefits and risks of using a PET-CT scanner are:

Benefits:

Any **one** from:

- Together the scans provide more accurate diagnoses than the two scans performed separately; [1 mark]
- One scan can fill in the gaps that the other scan might not pick up e.g. cell / metabolic activity from PET not seen on CT; [1 mark]
- Both scans can be performed at the same time, limiting the time spent in hospital / scanners; [1 mark]

Risks:

Any **one** from:

- Both use ionising radiation which can damage the patient's cells, so using both carries double the exposure and risk; [1 mark]
- Both require injections of substances which may cause adverse side effects; [1 mark]

[Total: 6 marks]

3a

(a) Deduce whether the annihilation and the creation of electron-positron pairs can occur:

List the known quantities:

- Maximum kinetic energy of β^+ from F1-18 = 0.63 MeV
- Maximum kinetic energy of β^+ from O-15 = 1.73 MeV
- Rest mass of an electron, $m_e = 9.11 \times 10^{-31} \text{ kg}$
- Speed of light, $c = 3.00 \times 10^8 \text{ m s}^{-1}$

Calculate the rest mass-energy of an electron-positron pair:

- Minimum possible photon energy: $E_{\min} = 2m_e c^2$
- $E_{\min} = 2 \times (9.11 \times 10^{-31}) \times (3.00 \times 10^8)^2 = 1.64 \times 10^{-13} \text{ J}$

- One scan can fill in the gaps that the other scan might not pick up e.g. cell / metabolic activity from PET not seen on CT; [1 mark]
- Both scans can be performed at the same time, limiting the time spent in hospital / scanners; [1 mark]

Risks:

Any **one** from:

- Both use ionising radiation which can damage the patient's cells, so using both carries double the exposure and risk; [1 mark]
- Both require injections of substances which may cause adverse side effects; [1 mark]

[Total: 6 marks]

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Calculate the rest mass-energy of an electron-positron pair:

- Minimum possible photon energy: $E_{\min} = 2m_e c^2$
- $E_{\min} = 2 \times (9.11 \times 10^{-31}) \times (3.00 \times 10^8)^2 = 1.64 \times 10^{-13} \text{ J}$
- Convert to MeV: $E_{\min} = \frac{1.64 \times 10^{-13}}{1.60 \times 10^{-19}} = 1.02 \text{ MeV}$ [1 mark]

Calculate the minimum possible energy of the annihilation photons:

- Min energy of one photon = $\frac{1}{2} \times 1.02 = 0.51 \text{ MeV}$

- (When O-15 / F-18 decay they) both emit a positron which annihilates with a nearby electron to produce two photons each with a minimum energy of 0.51 MeV; [1 mark]

Calculate the maximum possible energy of the annihilation photons from fluorine-18 decay:

- Maximum possible photon energy: $E_{max} = 2m_e c^2 + KE_{max}$
- Max energy available: $E_{max} = 1.02 + 0.63 = 1.65 \text{ MeV}$
- Max energy of one photon = $\frac{1}{2} \times 1.65 = 0.83 \text{ MeV}$
- $0.83 \text{ MeV} < 1.02 \text{ MeV}$ (therefore) photon (from F-18) does not have enough energy to create an electron-positron pair; [1 mark]

Calculate the maximum possible energy of the annihilation photons from oxygen-15 decay:

- Max energy available: $E_{max} = 1.02 + 1.73 = 2.75 \text{ MeV}$
- Max energy of one photon = $\frac{1}{2} \times 2.75 = 1.38 \text{ MeV}$
- $1.38 \text{ MeV} > 1.02 \text{ MeV}$ (therefore) photon (from O-15) has enough energy to create an electron-positron pair; [1 mark]

[Total: 4 marks]

3b

(b)

(i) Explain the observations of energy and momentum in detected gamma photons:

Observation 1: The majority of gamma photons have an energy of 511 keV and travel at exactly 180° to each other because...

- The majority of positrons dissipate all of their kinetic energy (before annihilation); [1 mark]
- By interacting with matter / atoms in the tissue of the patient; [1 mark]

Observation 2: With F-18 , a very small proportion of the gamma photons have energies greater than 511 keV and do not travel at exactly 180° to each other because...

Any **two** from:

- Emitted positrons (from fluorine-18) have non-zero kinetic energy (up to 0.63 MeV); [1 mark]
- Some of the positrons / electrons may not come to a complete rest (before annihilation); [1 mark]
- (Hence) the annihilation photons may have different energies / momenta; [1 mark]
- In order to conserve momentum, the photons may have different components of velocity (hence they would not move in exactly opposite directions); [1 mark]

Observation 3: With O-15 , a slightly larger proportion of the gamma photons have energies greater than 511 keV and do not travel at exactly 180° to each other because...

- Emitted positrons (from oxygen-15) can have higher kinetic energies than the positrons from fluorine-18; [1 mark]
- Hence, expected that a slightly larger proportion of these positrons would not come to a complete rest **OR** more positrons are expected to have non-zero energy / momentum when they annihilate; [1 mark]

(ii) If the majority of photons possessed energies greater than 511 keV:

Any **two** from:

- Many (photon pairs) would not travel in exactly opposite directions; [1 mark]
- This could make it harder to locate the exact position of the annihilation event; [1 mark]
- This could affect the overall image quality / resolution; [1 mark]

[Total: 8 marks]

You don't need to know the exact mechanisms by which positrons (or electrons) dissipate energy, you just need to be aware that they **lose energy** when they **interact with matter** as you have seen in various topics, such as in the photoelectric effect and the production of X-rays

3c

(c) Determine the number of photons produced during a PET scan using fluorine-18:

List the known quantities:

- Half-life, $t_{1/2} = 109.8 \text{ min}$
- % of decays that occur by β^+ emission = 97%
- Initial activity, $A_0 = 450 \text{ MBq} = 450 \times 10^6 \text{ Bq}$
- Time, $t = 30 \text{ min}$

Determine the decay constant of fluorine-18:

- Decay constant: $\lambda = \frac{\ln 2}{t_{1/2}}$
- $\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{109.8 \times 60} = 1.052 \times 10^{-4} \text{ s}^{-1}$ [1 mark]

Determine the initial number of fluorine-18 nuclei in the tracer:

- Activity: $A = \lambda N \Rightarrow N = \frac{A}{\lambda}$
- Initial number of nuclei: $N_0 = \frac{A_0}{\lambda} = \frac{450 \times 10^6}{1.052 \times 10^{-4}} = 4.28 \times 10^{12}$ [1 mark]

Determine the number of nuclei remaining after 30 minutes:

Determine the decay constant of fluorine-18:

- Decay constant: $\lambda = \frac{\ln 2}{t_{1/2}}$
- $\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{109.8 \times 60} = 1.052 \times 10^{-4} \text{ s}^{-1}$ [1 mark]

Determine the initial number of fluorine-18 nuclei in the tracer:

- Activity: $A = \lambda N \Rightarrow N = \frac{A}{\lambda}$
- Initial number of nuclei: $N_0 = \frac{A_0}{\lambda} = \frac{450 \times 10^6}{1.052 \times 10^{-4}} = 4.28 \times 10^{12}$ [1 mark]

Determine the number of nuclei remaining after 30 minutes:

- Radioactive decay: $N = N_0 e^{-\lambda t} \Rightarrow N = N_0 e^{-\frac{t \ln 2}{t_{1/2}}}$
- Remaining number of nuclei: $N = 0.97 \times (4.28 \times 10^{12}) \times e^{-\frac{30 \times \ln 2}{109.8}}$
- Remaining number of nuclei: $N = 3.44 \times 10^{12}$ [1 mark]

Determine the number of photons emitted:

- Each positron-electron annihilation corresponds to the production of two photons
- Therefore, number of photons = 2 × number of positrons emitted
- Number of photons produced: $\Delta N = 2 \times (4.28 - 3.44) \times 10^{12}$
- Number of photons produced = 1.68×10^{12} [1 mark]

[Total: 4 marks]

Make sure to multiply the number of nuclei remaining by 0.97! This is because 97% of the decays of fluorine-18 are beta-plus, or positron, decay and positrons are the only type of decay we are interested in

3d

(d) Determine the initial activity of oxygen-15 required:

List the known quantities:

- Half life of O-15, $t_{1/2} = 122 \text{ s}$
- Number of photons = 1.68×10^{12} (from (c))
- Number of positrons emitted, $N = 8.4 \times 10^{11}$
- % of decays that occur by β^+ emission = 100%
- Time, $t = 6 \text{ min} = 6 \times 60 = 360 \text{ s}$

Determine the decay constant of oxygen-15:

- Decay constant: $\lambda = \frac{\ln 2}{t_{1/2}}$
- $\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{122} = 5.682 \times 10^{-3} \text{ s}^{-1}$ [1 mark]

Determine the initial number of oxygen-15 nuclei in the tracer:

- Radioactive decay: $N = N_0 e^{-\lambda t} \Rightarrow N_0 = \frac{N}{e^{-\lambda t}}$
- Initial number of nuclei: $N_0 = \frac{8.4 \times 10^{11}}{e^{-(5.682 \times 10^{-3}) \times 360}}$
- Initial number of nuclei: $N_0 = 6.50 \times 10^{12}$ [1 mark]

Determine the initial activity of oxygen-15 required:

- Activity: $A = \lambda N$
- $A_0 = \lambda N_0 = (5.682 \times 10^{-3}) \times (6.50 \times 10^{12})$
- $A_0 = 3.7 \times 10^{10} \text{ Bq} = 37 \text{ GBq}$ [1 mark]

[Total: 3 marks]

